

SEMESTER-V

UME511: AUTOMOBILE ENGINEERING				
	L	T	P	Cr
	3	0	2*	3.5
Course Objectives: To deliver basic knowledge of different components of automobiles and expose the students with performance parameters of a vehicle. Course provides the learning of design procedure of various components and factors affecting operation of vehicle on road. Objective also involves the enhancement of fault diagnosis and troubleshooting capabilities.				
Syllabus: Introduction: Conventional motor vehicle, vehicle classification and layouts, frame and frameless construction, design requirements of different chassis types, vehicle dimensions, components of an automobile, power requirements, power and torque correlation. Clutch and Transmission: Automatic clutch, fluid coupling, torque convertor, requirements for manual and automatic transmission, their type and constructional detail, sliding mesh gearbox, constant mesh gearbox, synchromesh gearbox, gear ratios of planetary gear system, planetary gear selection mechanism, gear change mechanism, Simpson transmission, hydraulic gear change mechanism for automatic transmission, Constantly Variable Transmission System (CVT) construction and operation, Dual Clutch Transmission (DCT) construction and operation mechanism. Drive Line: Types of propeller shafts and support bearings, differential gear ratio, 4X4 drive, differential locks, limited slip differentials (LSD), types of axles and hubs. Wheel and Tyres: Types of tyres and their requirements, tyre construction, tubed and tubeless tyres, radial tyres, tyre design specifications and coding, tread pattern, speed ratings, tyre wear, disc pressed wheels, alloy wheels, split rims. Braking System: Introduction to braking system and their types, brake fade, stopping distance, hydraulic braking system, brake line layouts and safety features, solid disc and ventilated disc brakes, vacuum booster brakes, self-energizing brakes, antilock braking system (ABS). Suspension and Steering: Steering mechanisms and steering system including power steering, caster, camber, camber roll, steering axis inclination, toe-in, toe-out, rack and pinion steering, recirculating ball screw (RBS) steering, suspension principle, coil spring and leaf spring suspension, rigid axle suspension and independent suspension, types of dampers, Emission control devices: Catalytic convertor and its types, EGR. Trends in automobile sector: Hybrid, electric and solar powered vehicles.				
Laboratory Work: Study of vehicle chassis and construction, study of single plate and multi- plate clutch in an automobile, construction and working of following gear boxes: Contact mesh gear box;				

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synchronous gear box, parts of automatic transmission system, components of suspension system of automobile (2 wheel, 4 wheel), steering system of an automobile, starting system, braking system of an automobile, study of radiator, study of differential, axles, study of propeller shaft, universal joints and slip joint, study of catalytic convertor; Practical determination of the gearbox and rear axle ratios of a vehicle without dismantling any of these.

Course Learning Outcomes (CLOs):

The students will be able to:

1. demonstrate the power requirement of a vehicle under different operating conditions.
2. demonstrate the functioning of different power transfer mechanisms.
3. correlate the vehicle braking, steering, comfort requirements for the safety.
4. solve the technical issues related to vehicle design and malfunctioning of different components through fault-diagnosis and troubleshooting.

Text Books

1. Automobile Mechanics, Giri, N. K., Khanna Publishers, New Delhi, 2011.
2. Fundamentals of Motor Vehicle Technology, Hiller, V. A. W., Nelson Thornes, UK, 2012.

Reference Books

- 1 The Motor Vehicle, Garrett, T. K., Newton, K. and Steeds, W., Butterworth-Heinemann, Great Britain, London, 2001.
- 2 A. A., Book of the Car, Automobile Association, Norton, London, 1977.
- 3 Advance Vehicle Technology, Heinz, H., Arnold Publishers, Butterworth-Heinemann, London, 1999.
- 4 Automotive Mechanics, Crouse, W. and Anglin, D., Tata McGraw Hill, New Delhi 2006.
- 5 Engine and Vehicle Technology, Heinz, H, Arnold Publishers, Butterworth-Heinemann, London, 2002.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Quizzes/Lab Evaluations)	35

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